

The Wizard thinks he finally has mastered some of the spells in his book of magic. He needs numerical alignment in the stars to do it though. Help the wizard align the numbers in his spell book evenly.

Input

You will receive a positive number on a single line, from 4 to 10 digits in length. One digit will repeat more than the others.

```
3334233
```



Output

Check to see if there is enough value available in the higher columns to redistribute to the lower columns to make each column contain the same digit. The original digit repeated the most is the digit all columns should match. If it is possible, output 4 lines: the ORIGINAL number, a row showing values SUBTRACTED, a row showing values ADDED, and the FINAL row.

If there is either too much or not enough value in higher columns to even out all of the digits, also output 4 lines: the ORIGINAL number, how much is AVAILABLE to subtract, how much is NEEDED to add, then the target number IMPOSSIBLE to reach.

```
3334233 ORIGINAL
0001000 SUBTRACTED
0000100 ADDED
3333333 FINAL
```

Discussion

In Input 1, 3 is the most repeated. There is one column of 2, and one column of 4, so the redistribution is easy. Simply take one from the '4' column, and add it to the '2' column.

In Input 2, 8 is the most repeated. However, there are not enough high columns to make the others equal 8.

The input number will never have all digits the same.

Additional Examples

Input 2	Output 2
887971878	887971878 ORIGINAL 000100000 AVAILABLE 001017010 NEEDED 888888888 IMPOSSIBLE